SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

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1. Experiment Title:

Develop web applications using JavaScript to reverse a string and count the number of vowels in a paragraph using string methods and regular expressions.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Case Study 6.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Experiment 6 - Case Studies</title>
    <link rel="stylesheet" href="Case Study 6.css">
</head>
<body>
    <div class="container">
        <h2>Experiment 6 - Case Studies</h2>

        <div class="box">
            <h3>Case Study 1: Reverse a String</h3>
            <label for="strInput"><b>Enter a String:</b></label><br>
            <input type="text" id="strInput" placeholder="Enter string to reverse">
            <br>
            <button onclick="reverseString()">Reverse String</button>
            <div id="result"></div>
        </div>

        <div class="box">
            <h3>Case Study 2: Vowel Counter in Paragraph</h3>
            <label for="paragraphInput"><b>Enter a Paragraph:</b></label><br>
            <textarea id="paragraphInput" rows="4" placeholder="Enter a paragraph to count
vowels..."></textarea>
            <br>
            <button onclick="countVowels()">Count Vowels</button>
            <div id="vowelResult"></div>
        </div>

        <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-
top: 1px solid lightgray; padding-top: 15px;">
            Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Case Study 6.js"></script>
</body>
</html>
```

Case Study 6.css

```
body {
    font-family: Arial, sans-serif;
    margin: 40px;
    background-color: whitesmoke;
}

h2 {
    text-align: center;
    color: darkblue;
}

h3 {
    color: darkblue;
    margin-top: 0;
    margin-bottom: 15px;
    border-bottom: 1px solid #eee;
    padding-bottom: 5px;
}

.container {
    max-width: 650px;
    margin: 0 auto;
    background: white;
    padding: 20px;
    border-radius: 8px;
    border: 1px solid lightgray;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
}

.box {
    border: 1px solid #ccc;
    padding: 20px;
    margin-top: 20px;
    border-radius: 8px;
    background-color: #fafafa;
}

input, textarea, button {
    padding: 10px;
    margin: 10px 0;
    width: 95%;
    font-family: inherit;
    box-sizing: border-box;
}

textarea {
    resize: vertical;
}

button {
    width: 150px;
    background-color: royalblue;
    color: white;
    border: none;
    border-radius: 4px;
    cursor: pointer;
    font-weight: bold;
}

button:hover {
    background-color: blue;
}

#result, #vowelResult {
    margin-top: 15px;
    padding: 15px;
    background-color: #ffffff;
    border: 1px solid #ddd;
```

```
    border-radius: 6px;
    font-size: 15px;
}
```

Case Study 6.js

```
function reverseString() {
    let str = document.getElementById("strInput").value;

    if (str.trim() === "") {
        document.getElementById("result").innerHTML = "<b style='color:red;'>Please enter a
string!</b>";
        return;
    }

    let reversedStr = str.split("").reverse().join("");

    document.getElementById("result").innerHTML = `
        <p><b>Original String:</b> ${str}</p>
        <p><b>Reversed String:</b> ${reversedStr}</p>
    `;
}

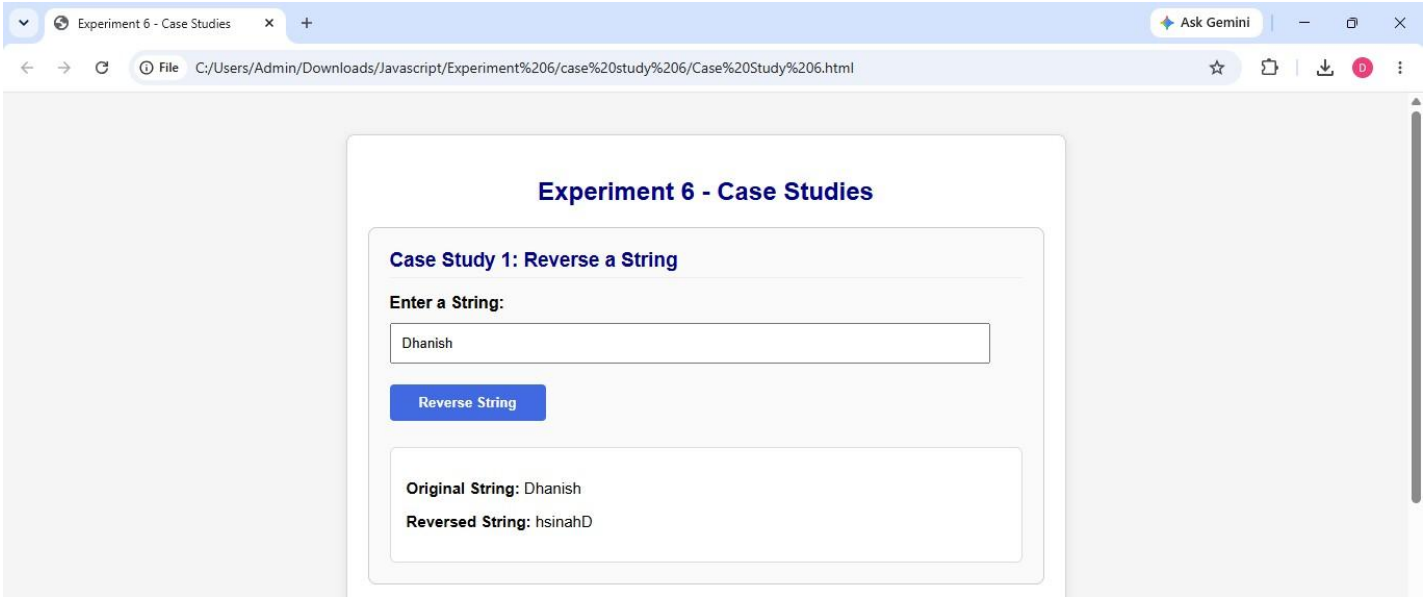
function countVowels() {
    let paragraph = document.getElementById("paragraphInput").value;

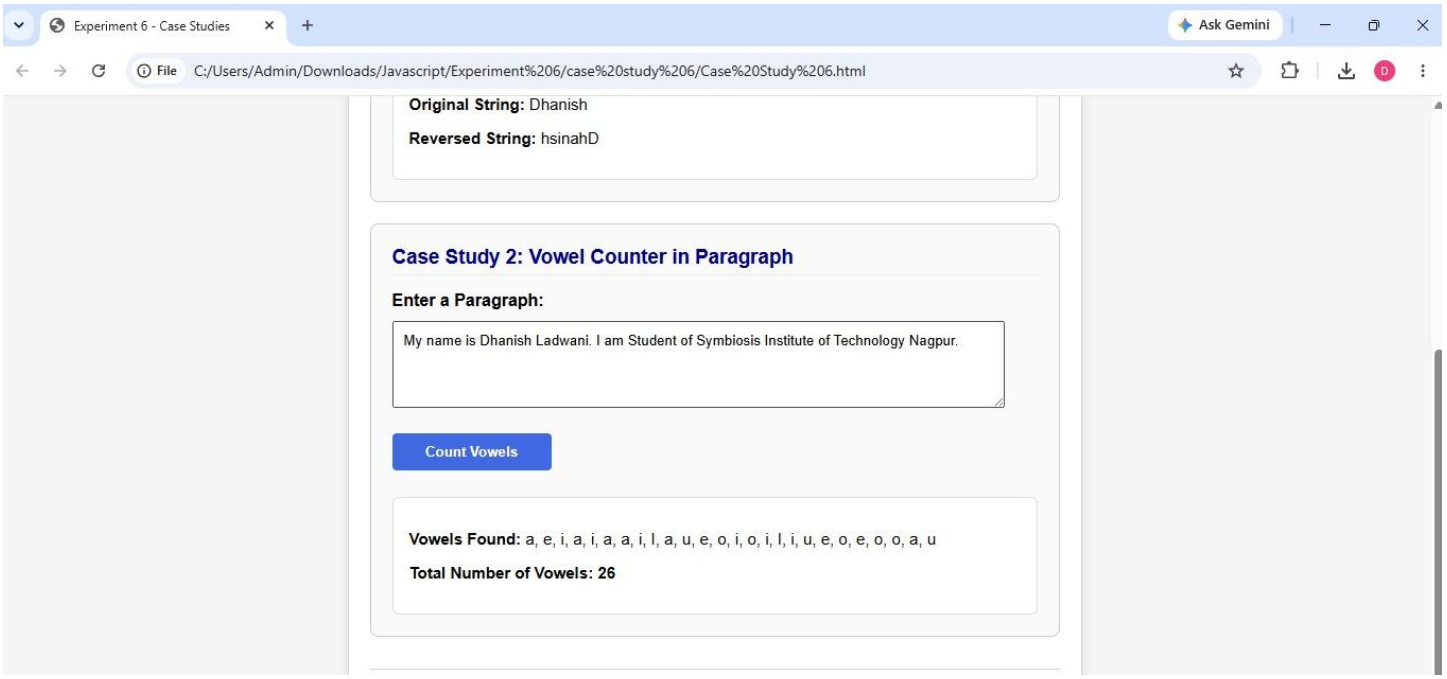
    if (paragraph.trim() === "") {
        document.getElementById("vowelResult").innerHTML = "<b style='color:red;'>Please enter a
paragraph!</b>";
        return;
    }

    let vowels = paragraph.match(/[aeiou]/gi);
    let vowelCount = vowels ? vowels.length : 0;
    let vowelsList = vowels ? vowels.join(", ") : "No vowels found";

    document.getElementById("vowelResult").innerHTML = `
        <p><b>Vowels Found:</b> ${vowelsList}</p>
        <p><b>Total Number of Vowels:</b> <b>${vowelCount}</b></p>
    `;
}
```

4. Output:





5. Result/Conclusion:

Two small case studies were implemented to reinforce string handling in JavaScript. The first uses the split(""), reverse() and join("") chain to reverse a user-entered string, with a validation check for empty input. The second uses the match() method with a regular expression to find all vowels in a paragraph, displaying the matched vowels and their total count. Both case studies reinforced core string methods and simple regular expression pattern matching for text processing in JavaScript.